

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 3343	Course Title: Data Structure
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- Students will be able to demonstrate different linear and nonlinear data structures and develop skills to identify and apply the suitable data structure for the given real world problem.

Course Prerequisites:

Topic	Course name	Semester
Control Structures, Arrays, Functions	Problem solving and Programming	2

Course Outcomes

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the fundamental concepts of linear data structures stack and queue	10	Applying
CO2	Illustrate searching and sorting algorithms and nonlinear data structure linked list	12	Applying
CO3	Demonstrate non linear data structures - Trees	11	Applying
CO4	Demonstrate non linear data structures - Graphs	10	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the fundamental concepts of linear data structures stack and queue		
M1.01	Explain the concepts of structures, pointers and files in C	2	Understanding
M1.02	Describe data structure and their classification	1	Understanding
M1.03	Illustrate Array based sequences & operations	2	Applying
M1.04	Demonstrate Array Based Stack ADT operations	2	Applying
M1.05	Use stack operations to convert infix expressions to postfix	1	Applying
M1.06	Demonstrate Array Based Queue ADT operations	2	Applying

Contents:

Advanced C concepts: Concept of Pointers, structures

Data Structures: Definition, Classification- Linear Data Structures, Non-Linear Data Structures, Abstract Data Type(ADT)

Arrays: Array based sequences, operations- Traverse, insert, delete, search, update

Stack: Introduction to Stack ADT - Array Representation of Stacks - Operations on a Stack - Applications of Stacks - Infix-to-Postfix Conversion.

Queues: Introduction to Queue ADT - Array Representation of Queues - Operations on a Queue

CO2	Illustrate searching and sorting algorithms and nonlinear data structure linked list		
M2.01	Outline the concept of Linked List and its representation in memory.	2	Understanding
M2.02	Implement the operations of a Singly Linked List	2	Applying
M2.03	Implement stack ADT using singly Linked list	2	Applying
M2.04	Implement queue ADT using singly Linked list	2	Applying
M2.05	Demonstrate Linear Search algorithm	1	Applying
M2.06	Illustrate Binary Search algorithm	1	Applying
M2.07	Use Bubble Sort algorithm	1	Applying
M2.08	Sketch Quick Sort algorithm	1	Applying
	Series Test – I	1	
Contents: Linked Lists: Singly Linked List - Representation in Memory, Operations on a Singly Linked List – Insertion, Deletion, Searching and Traversal., Linked List Representation of Stack and Queue Searching algorithms: Linear Search, Binary Search. Sorting Algorithms: Bubble Sort, Quick Sort			
CO3	Demonstrate non linear data structures - Trees		
M3.01	Describe the terminologies related to Binary Trees	1	Understanding
M3.02	Classify different types of Binary Trees	1	Understanding
M3.03	Describe the representation of a Binary Tree	1	Understanding
M3.04	Demonstrate inorder tree traversal algorithm	2	Applying
M3.05	Illustrate preorder tree traversal algorithm	1	Applying
M3.06	Demonstrate postorder tree traversal algorithm	1	Applying
M3.07	Demonstrate a Binary Search Tree with different operations like Creation, Insertion, Search and Deletion.	4	Applying

Contents:

Trees: Definition - Basic Terminologies - Node, Parent, Child, Link, Root, Leaf, Level, Height of a tree and node, Depth of a tree and node, Degree of a tree and node, sibling, Ancestors, Path, Path Length. Binary Tree-

Types of Binary Trees: Full, Complete

Representations of a Binary Tree: using Linked List

Traversal Algorithms- Inorder, Preorder, Postorder.

Binary Search Tree: Creation-Insertion –Traversal – Search – Deletion

CO4	Demonstrate non linear data structures - Graphs		
M4.01	Describe the terminologies related to Graphs	1	Understanding
M4.02	Classify different types of Graphs	1	Understanding
M4.03	Explain the different representation of a Graph	2	Understanding
M4.04	Illustrate Graph Traversal algorithms.	4	Applying
M4.05	Explain Warshall's algorithm to find the shortest path.	2	Applying
	Series Test – II	1	

Contents:

Graphs: Terminologies - Vertex, Edge, Adjacent vertices, Self-loop, Parallel edges, Isolated vertex, Degree of vertex, Pendant vertex, Subgraph, Paths and Cycles.

Types of Graphs - Directed, Undirected, Simple, Complete, Cyclic, Acyclic, Connected.

Representation of Graphs - Set – Linked – Matrix

Graph Traversals -BFS, DFS

Text / Reference:

T/R	Book Title / Author
T1	Data Structures using C, E. Balagurusamy, Tata Mcgraw Hill
R1	Reema Thareja, Data Structures Using C, Oxford University Press India.
R2	Richard F. Gilberg, Behrouz A. Forouzan, Data Structures: A Pseudocode approach with C, 2nd ed., Cengage Learning, India, 2005

Online Resources:

Sl. No.	Website Link
1	https://nptel.ac.in/courses/106103069
2	http://www.tutorialspoint.com/cprogramming/
3	https://www.programmingsimplified.com/c/data-structures
4	https://www.programiz.com/dsa